

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 313 — Past Questions & Materials
by Renance

Course: MTH 313
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

This study pack was compiled and restructured by Renance for MTH 313 students at Federal University of Petroleum Resources, Effurun. It brings together 2 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 2 — MTH 313 or 303 Complex Analysis 1 2018 2019 First Semester

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE =

x B.Sc (MATHEMATICS) DEGREE SEMESTER EXAMINATION (2018/2019), FIRST SEMESTER. |

= aa #07H313 (Complex Anatraa April, 2019. Vime:2hrs. _ =. ___-__—

Instruction: Answer ALL questions | aie" 3

All Questions Carry Equal Marks . Se

Question 1. ' ny | , k

o (a) Define a derivable (or diffe tiable) function f at Z_p), hence or otherwise state the Cauchy- | Riemann identities for a differentiable/analytic function $w = u + iv$ | ?

i. (b) Prove that $\sinh z = -i \sin(iz)$ z

S (c) Evaluate $\lim_{x \rightarrow 0} (2xy - 4x^2 + iy^2)$ | aT ae

Question 2. $\sim | > | , Zz$.

6 (a) If $z = (2 + i)^{31}$? express Zz in polar form, hence find radius r and arg 8 of 22 a9

ch (b) Let $z_1, z_2 \in \mathbb{C}$, -prove that $27 > 2p = j * Z, | ae$

by (c) Represent in the plane the set of complex numbers z such that $|z - 2i| = |z - 1|$. - fa

Quetien 3 = = #2 ee

A. (a) Define an analytic function, write down two examples =

A, Ab) Let If $f(z) =$ = detaining whe point at which $f(z)$ is not analytic. : | | | z

Q (c) Prove that the Sunetion f-s 4 "such that wane hy | = oo Pea tr'

| ate arts $G + \frac{2}{x}$

is differentiable on $\mathbb{C} - \{-i, f\}$.. | =: he.

Question 4 "| - e ' oe Ge —s

Sh (a) Evaluate $\sinh 2i$? Kaas oe Oe ar

b (b) Find all the values of z such that $\sin z = 2$ LL

— ame eee aaline qe *

"a (c) Compute the derivatives f al the analytic functions, e" and. justify your answers. fa

Question5 —

©),,,{a) Define a Bilinear Transformation. _ oA "—

2 (b) Let $T(z)$ and $s(z)$ be Bilinear transformations, show that the composition $(T \circ s)(z)$ is also a ~ ~ »

Bilinear Transformation. — 7 =F

5 (c) Find the bilinear transformation which map $1, i, -1$ onto $1 + i, 0, 1 - i$ respectively. oo,

we

Question 6

\$3 (a) What is a fixed point of a complex function f

> (b) Given that $T(z) = (3z + 2y/z - 5)$ is a bilinear transformation, find all the fixed points of T -

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a (c) Find the image of the circle $|z| = r, r \neq 0$ under the bilinear transformation

$7, - \sim go-i$

+ $zt+f$

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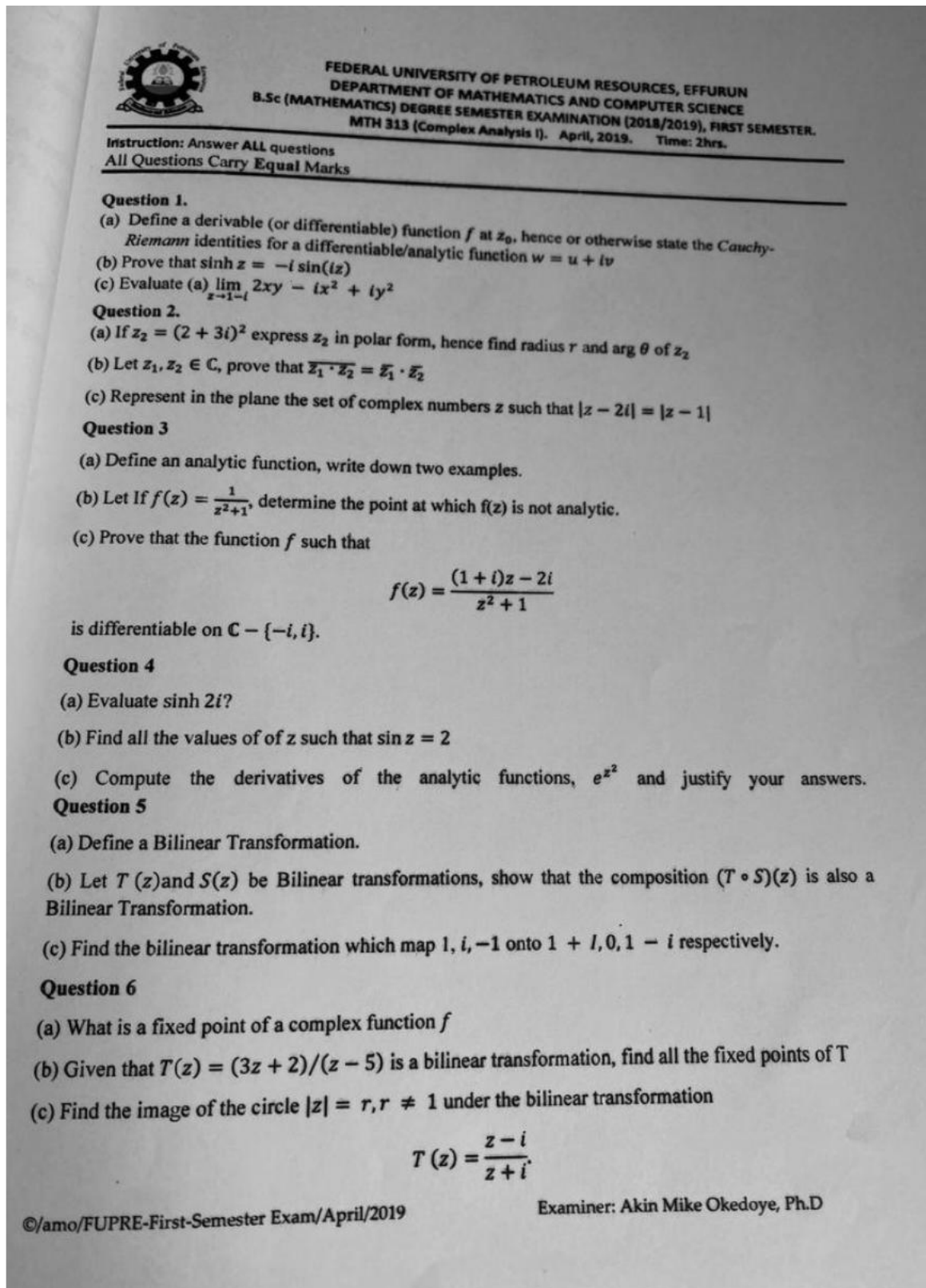
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Paper 2 of 2 — MTH 313, 2019

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
B.Sc (MATHEMATICS) DEGREE SEMESTER EXAMINATION (2018/2019), FIRST SEMESTER.
MTH 313 (Complex Analysis I). April, 2019. Time: 2hrs.

Instruction: Answer ALL questions
All Questions Carry Equal Marks

Question 1.

- Define a derivable (or differentiable) function f at z_0 , hence or otherwise state the *Cauchy-Riemann* identities for a differentiable/analytic function $w = u + iv$
- Prove that $\sinh z = -i \sin(iz)$
- Evaluate (a) $\lim_{z \rightarrow 1-i} 2xy - ix^2 + iy^2$

Question 2.

- If $z_2 = (2 + 3i)^2$ express z_2 in polar form, hence find radius r and arg θ of z_2
- Let $z_1, z_2 \in \mathbb{C}$, prove that $\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}$
- Represent in the plane the set of complex numbers z such that $|z - 2i| = |z - 1|$

Question 3

- Define an analytic function, write down two examples.
- Let If $f(z) = \frac{1}{z^2 + 1}$, determine the point at which $f(z)$ is not analytic.
- Prove that the function f such that

$$f(z) = \frac{(1+i)z - 2i}{z^2 + 1}$$

is differentiable on $\mathbb{C} - \{-i, i\}$.

Question 4

- Evaluate $\sinh 2i$
- Find all the values of z such that $\sin z = 2$
- Compute the derivatives of the analytic functions, e^{z^2} and justify your answers.

Question 5

- Define a Bilinear Transformation.
- Let $T(z)$ and $S(z)$ be Bilinear transformations, show that the composition $(T \circ S)(z)$ is also a Bilinear Transformation.
- Find the bilinear transformation which map $1, i, -1$ onto $1 + i, 0, 1 - i$ respectively.

Question 6

- What is a fixed point of a complex function f
- Given that $T(z) = (3z + 2)/(z - 5)$ is a bilinear transformation, find all the fixed points of T
- Find the image of the circle $|z| = r, r \neq 1$ under the bilinear transformation

$$T(z) = \frac{z - i}{z + i}$$

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